CS 411W: Professional Workforce Development II

Lab 1: Project Outline

FAÍCT: Fact AI Checking Tool

Version 1

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Table of Contents

1. Introduction 3

2. Product Description 3

2.1 Key Product Features and Capabilities 4

2.2 Major Functional Components 5

3. Identification of Case Study 6

3.1 Case Study Participants and Scenario 6

3.2 Prototype Demonstration 7

3.3 Full-Capability Implementation 7

3.4 Broader User and Customer Base 7

4. Prototype Description 8

4.1 Prototype Scope 8

4.2 Prototype Workflow 8

4.3 Limitations and Evaluation 9

5. Glossary 9

6. References 10

Table of Figures: Figure 1. Major Functional Components Diagram

Table of Tables: Table 1. Prototype Feature Scope

1. Introduction

Students frequently encounter claims about financial aid through social media, group chats, search results, and conversations with friends. A post may state that withdrawing from a class automatically cancels financial aid or that every student qualifies for a particular grant. These claims can sound convincing while leaving out important details about enrollment status, eligibility, institutional policies, or individual circumstances. Acting on incomplete information could affect a student's academic and financial decisions.

Verifying such claims often means reading information across several websites, deciding which sources apply, and identifying conditions that change the answer. Official information from Federal Student Aid and a student's university may address different parts of the same question. A general internet search does not always make those distinctions clear, especially when a claim is framed as a simple yes-or-no statement.

FAÍCT, the Fact AI Checking Tool, is a proposed evidence-based claim verification system. Its purpose is to help users evaluate claims by locating relevant sources, presenting a clear assessment, and explaining what the available evidence supports. Rather than asking users to trust an unexplained AI judgment, FAÍCT is designed to display source links, short explanations, and uncertainty when a statement cannot be verified with the information available.

The initial case study focuses on college students checking financial aid claims, with Old Dominion University (ODU) students serving as the sample user group. This narrower setting gives the prototype a specific problem to demonstrate while preserving the long-term goal of a tool that can evaluate claims in other subjects. The prototype is intended to support decision-making, not replace a university financial aid office or provide a binding eligibility determination.

2. Product Description

FAÍCT is a browser-based software product. A user enters a claim in a web interface, submits it for analysis, and receives an organized result that includes an assessment, a brief explanation, and links to relevant evidence. For the financial aid case study, the system gives priority to primary sources such as StudentAid.gov and published ODU financial aid guidance when they are available and relevant. Source information should be distinguishable from the system's own interpretation.

The proposed system consists of a frontend interface, backend service, claim-analysis process, evidence retrieval component, and a results display. It can operate through a standard internet-connected computer without specialized user hardware. External information services may be used to retrieve evidence, and a database may be added where storage is necessary. The exact technologies and integrations are implementation choices rather than promises of completed functionality.

2.1 Key Product Features and Capabilities

2.1.1 Claim Submission

Users enter a short written claim, such as "If I withdraw from one class, I will lose all of my financial aid." The interface submits the text to the backend and indicates when analysis is in progress. Text entry is the primary submission method for the prototype.

2.1.2 Claim Analysis and Evidence Retrieval

FAÍCT identifies the statement being evaluated and retrieves sources that may support, contradict, or qualify it. Financial aid policies frequently depend on conditions, so the system should recognize when a broad claim cannot be answered without further context, such as a student's enrollment level, aid program, or school requirements.

2.1.3 Credibility Assessment

The system provides an understandable assessment, such as supported, contradicted, mixed or conditional, or insufficient evidence. A result is not presented as absolute truth. The assessment is tied to the retrieved evidence and can change when authoritative policies or the relevant facts change.

2.1.4 Explanation and Source Display

Results include a plain-language explanation and identifiable source links so users can review the original material. Where information is missing, FAÍCT should say what the user needs to confirm rather than invent an answer. This is especially important for decisions involving financial aid.

2.1.5 Future Media and Accessibility Features

A full version may include speech-to-text, video transcripts, browser integration, and other accessible ways to submit claims. These are expansion possibilities and are not required for the initial December prototype. Similarly, detecting whether content was generated by AI is a separate problem from verifying whether a particular claim is accurate and is not part of the central case-study workflow.

2.2 Major Functional Components

Frontend Interface

Accepts written claims, presents loading and error states, and shows an understandable result with source links.

Backend/API Layer

Receives submissions, validates input, coordinates analysis and evidence requests, and sends a structured response to the frontend.

Claim Analysis

Interprets the claim and organizes the evidence into a qualified assessment with an explanation; it must allow an insufficient-evidence outcome.

Evidence Retrieval and Source Evaluation

Searches or accesses relevant publications, records the source titles and URLs, and favors authoritative sources when appropriate.

Results Presentation

Separates the assessment, explanation, supporting or conflicting evidence, and suggested next verification step.

Optional Data Storage

May retain limited application data needed for development or history features, subject to privacy safeguards; the case-study workflow does not require student financial records.

Figure 1. Major Functional Components Diagram

Input and coordination	Analysis and evidence	Output
Student enters a text claim ↓ Frontend interface ↓ Backend/API	Claim analysis ↔ Evidence retrieval from relevant sources	Assessment and explanation ↓ Source links and conditions ↓ Student reviews next steps

Figure 1 shows the logical flow of a claim from the user interface through backend coordination, analysis, evidence gathering, and return of a reviewable result. It is a functional model; individual services may be combined during prototype development.

3. Identification of Case Study

The case study follows four ODU students who encounter conflicting information about how dropping or withdrawing from a class might affect financial aid. The students represent the initial user group, and the scenario reflects a real decision that may require comparing federal aid rules with university guidance. The participants do not need to disclose account credentials, student identification numbers, or personal financial aid records to use the prototype.

3.1 Case Study Participants and Scenario

During a study session, one student sees a social media post claiming, “Withdrawing from one class automatically means you lose all of your financial aid.” Another student has heard a different explanation. Rather than treating either statement as definitive, the group submits the claim to FAÍCT. A student using the tool wants to know whether the statement is broadly accurate, which details might change the outcome, and where the information can be confirmed.

FAÍCT displays a qualified assessment and evidence from relevant sources. It explains that a student’s outcome may depend on the type of aid, enrollment status, timing of the withdrawal, and applicable institutional or federal rules. Instead of promising a personal decision, the system advises the student to check the applicable official guidance and contact the financial aid office for circumstances not established by the claim. Other participants review the linked sources and discuss whether the result is understandable.

3.2 Prototype Demonstration

The demonstrator enters the sample claim, initiates analysis, and views a result containing a credibility category, a short explanation, and source references. The demonstration emphasizes whether a student can follow the reasoning and recognize that financial aid claims often depend on conditions. Where the retrieved evidence is incomplete, the system should explicitly acknowledge that limitation.

3.3 Full-Capability Implementation

A future version could accept audio, video, and web links; provide more extensive source comparisons; support saved histories; and extend its source coverage beyond financial aid. These additions would require further development, testing, privacy decisions, and ongoing source maintenance. They are separate from the working prototype used for the case study.

3.4 Broader User and Customer Base

The immediate audience is college students who encounter financial aid information online. In the longer term, FAÍCT could support students checking other campus policies, researchers verifying claims, and members of the public evaluating information in other subject areas. The broader product would use the same evidence-first workflow, but each new subject would require suitable sources and evaluation criteria.

4. Prototype Description

The initial prototype demonstrates the most important user interaction: moving from a questionable written claim to a transparent, source-supported answer. It is intentionally narrower than the full product to keep the development goals attainable and testable.

4.1 Prototype Scope

Feature	Prototype commitment
Written claim input	Core: student submits one short text claim
Backend processing	Core: request is received and a result is returned
Evidence retrieval	Core target: retrieve or present relevant official source material for the selected scenario
Assessment and explanation	Core: display a qualified result and clear rationale
Source citations	Core: present source titles and navigable links
Conditional or uncertain result	Core: identify missing context instead of overclaiming
Speech-to-text, video, AI-content detection	Future scope: not required for this prototype
Accounts, browser extension, general-topic coverage	Future scope: not required for this prototype

Table 1. Prototype Feature Scope

4.2 Prototype Workflow

1. A student enters a financial aid claim in the browser interface.

2. The frontend sends the text to the backend service.

3. The system finds relevant evidence and prepares a qualified assessment.

4. The user receives an explanation with source titles and links.

5. The student reviews any missing conditions and determines what to verify with an official source or financial aid counselor.

4.3 Limitations and Evaluation

The prototype will not calculate individual aid eligibility, access private student records, or guarantee that retrieved policies are complete or current. It should avoid treating every financial

aid rule as universal. An unsupported or conditional result is preferable to a confident but misleading answer. Evaluation can focus on whether the sample workflow completes, whether citations open the expected sources, whether conditional language is understandable, and whether students can identify a useful next step after reading the response.

5. Glossary

AI: Artificial intelligence; computer-based systems used for tasks such as language processing.

API: Application Programming Interface; a mechanism for software components to communicate.

Backend: The server-side service that receives and processes application requests.

Claim: A statement submitted for evaluation against evidence.

Conditional Claim: A statement whose accuracy depends on circumstances or additional facts.

Credibility Assessment: A qualified description of how available evidence relates to a claim.

Evidence Retrieval: The process of finding relevant source material for claim evaluation.

FAÍCT: Fact AI Checking Tool.

Frontend: The browser interface used to submit claims and view results.

Misinformation: False or misleading information, regardless of intent.

Primary Source: A direct or authoritative source of information about the rule or event under review.

Prototype: A limited implementation that demonstrates selected functions.

Speech-to-Text: Technology that transcribes spoken audio into written text.

6. References

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